

Machine Learning Interview Questions

Questions Only

1. ML Fundamentals

1. What is Machine Learning?
2. What are the main types of Machine Learning?
3. What is supervised learning?
4. What is unsupervised learning?
5. What is reinforcement learning?
6. What is the difference between AI, ML, and Deep Learning?
7. What is a Machine Learning model?
8. What is a feature?
9. What is a target variable?
10. What is training data?
11. What is testing data?
12. What is validation data?
13. What is a training set and why is it required?
14. What is the difference between a parameter and a hyperparameter?
15. What is a baseline model?

2. Data Preprocessing

16. What is data preprocessing?
17. How do you handle missing values?
18. What is imputation?
19. When would you use mean, median, or mode imputation?
20. How do you handle duplicate records?
21. How do you handle outliers?
22. What is the IQR method?
23. What is the Z-score method?
24. What is feature scaling?
25. What is normalization?
26. What is standardization?
27. What is the difference between normalization and standardization?

- 28. What is label encoding?
- 29. What is one-hot encoding?
- 30. When should you use one-hot encoding?
- 31. What is ordinal encoding?
- 32. What is data leakage?
- 33. How can data leakage be prevented?

3. Feature Engineering and Selection

- 34. What is feature engineering?
- 35. Why is feature engineering important?
- 36. What is feature selection?
- 37. What is feature extraction?
- 38. What is dimensionality reduction?
- 39. What is correlation?
- 40. What is covariance?
- 41. What is multicollinearity?
- 42. What is VIF?
- 43. How can multicollinearity be handled?
- 44. What is PCA?
- 45. What is explained variance ratio?
- 46. What is SelectKBest?
- 47. What is the difference between feature selection and dimensionality reduction?

4. Regression

- 48. What is regression?
- 49. What is Linear Regression?
- 50. What are the assumptions of Linear Regression?
- 51. What is Multiple Linear Regression?
- 52. What is the difference between simple and multiple linear regression?
- 53. What is the cost function in Linear Regression?
- 54. What is gradient descent?
- 55. What is R-squared?
- 56. What is adjusted R-squared?

- 57. What are MAE, MSE, and RMSE?
- 58. Which regression metric would you choose and why?
- 59. What is Ridge Regression?
- 60. What is Lasso Regression?
- 61. What is Elastic Net Regression?
- 62. What is regularization?
- 63. What is the difference between L1 and L2 regularization?

5. Classification

- 64. What is classification?
- 65. What is binary classification?
- 66. What is multiclass classification?
- 67. What is Logistic Regression?
- 68. What is the sigmoid function?
- 69. What is a Decision Tree?
- 70. What are Gini Impurity and Entropy?
- 71. What is Information Gain?
- 72. What is Random Forest?
- 73. What is KNN?
- 74. What is SVM?
- 75. What is Naive Bayes?
- 76. What is the difference between KNN and K-Means?

6. Clustering

- 77. What is clustering?
- 78. What is K-Means clustering?
- 79. How does K-Means work?
- 80. How do you choose the value of K?
- 81. What is the elbow method?
- 82. What is silhouette score?
- 83. What are the limitations of K-Means?
- 84. What is hierarchical clustering?
- 85. What is DBSCAN?

86. What is the difference between K-Means and DBSCAN?

7. Model Evaluation

87. What is a confusion matrix?

88. What is accuracy?

89. What is precision?

90. What is recall?

91. What is specificity?

92. What is F1-score?

93. What is ROC-AUC?

94. When can accuracy be misleading?

95. What is cross-validation?

96. What is k-fold cross-validation?

97. What is stratified cross-validation?

98. What is overfitting?

99. What is underfitting?

100. What is the bias-variance tradeoff?

101. How can you reduce overfitting?

8. Ensemble Learning and Optimization

102. What is ensemble learning?

103. What is bagging?

104. What is boosting?

105. What is Random Forest?

106. What is Gradient Boosting?

107. What is XGBoost?

108. What is AdaBoost?

109. What is stacking?

110. What is hyperparameter tuning?

111. What is GridSearchCV?

112. What is RandomizedSearchCV?

113. What is early stopping?

9. ML Libraries and Pipelines

- 114. What is Scikit-learn?
- 115. What is a Pipeline in Scikit-learn?
- 116. What is ColumnTransformer?
- 117. What is SimpleImputer?
- 118. What is StandardScaler?
- 119. What is OneHotEncoder?
- 120. Why should preprocessing be included inside a pipeline?
- 121. What is model persistence?
- 122. What is joblib?
- 123. How do you save and load a trained ML model?